ML and DS Internship Practical

Month 1 – Python Fundamentals

ATM Simulation System

Submitted By:

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Submitted To: -

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Technology Used: Python

Development Environment: Visual Studio Code

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1 Objective: -

The objective of this project is to develop a console-based ATM Simulation System using Python. The application simulates basic banking operations such as deposit, withdrawal, balance inquiry, and transaction history. This project helps in understanding conditional statements, functions, and exception handling.

2 Introduction: -

An Automated Teller Machine (ATM) allows users to perform banking transactions without visiting a bank branch. This ATM Simulation project provides a simplified version of ATM functionality through a menu-driven interface. Users can deposit money, withdraw money, check their account balance, and view transaction history.

The project is designed to strengthen programming skills and demonstrate the practical implementation of Python fundamentals.

3 Features: -

1. Deposit Money

Users can deposit a specified amount into their account. The account balance is updated automatically after a successful transaction.

2. Withdraw Money

Users can withdraw money from their account. The system checks whether sufficient balance is available before processing the withdrawal.

3. Balance Inquiry

Users can view their current account balance at any time.

4. Transaction History

The system stores and displays all successful deposits and withdrawals performed during program execution.

5. Menu-Driven Interface

A user-friendly menu allows users to select different banking operations easily.

6. Exception Handling

Exception handling is used to manage invalid user input and prevent program crashes.

4 Concepts Used: -

Variables

- Used to store balance, transaction details, and user choices.

Data Types

- Integer
- Float
- String
- List

Conditional Statements

- Used for decision-making:
- Checking sufficient balance
- Validating user input
- Processing menu choices

Loops

- A while loop is used to repeatedly display the ATM menu until the user exits

Functions

- Separate functions are used for:
- Deposit
- Withdraw
- Balance Inquiry
- Transaction History

Lists

- Transaction records are stored in a list for future reference.

Exception Handling

- Used to handle:
- Invalid numeric input
- Unexpected user entries
- Runtime errors
- Delete → Delete Student Record

5 Project Workflow: -

1. Start the ATM Simulation program.
2. Display the main menu.
3. User selects an operation.
4. System processes the selected operation.
5. Balance and transaction history are updated.
6. Menu is displayed again.
7. User exits the application.

6 Learning Outcomes: -

Through this project, the following concepts were learned:

- Conditional Statements
- Functions
- Loops
- Lists
- Exception Handling
- Input Validation
- Menu-Driven Programming
- Transaction Management
- Problem Solving Skills

7 Advantages: -

- Easy to use
- Beginner-friendly
- Demonstrates banking transaction logic
- Improves programming skills
- Enhances logical thinking and decision-making

8 Limitations: -

- Data is stored temporarily during program execution.
- Transactions are not saved permanently.
- No database integration.

9 Testing Report: -

Test ID	Test Description	Test Input	Expected Result	Status
TC01	Deposit valid amount	500	Amount deposited successfully	Pass

TC02	Deposit large amount	10000	Amount deposited successfully	Pass
TC03	Deposit zero amount	0	Validation message displayed	Pass
TC04	Deposit negative amount	-500	Error message displayed	Pass
TC05	Withdraw valid amount	200	Amount withdrawn successfully	Pass
TC06	Withdraw amount equal to balance	Full Balance	Withdrawal successful	Pass
TC07	Withdraw amount greater than balance	Balance + 100	Insufficient balance message displayed	Pass
TC08	Withdraw negative amount	-100	Error message displayed	Pass
TC09	Withdraw zero amount	0	Validation message displayed	Pass
TC10	Check account balance	Balance Inquiry	Current balance displayed	Pass
TC11	View transaction history after deposit	Deposit Transaction	Transaction displayed	Pass
TC12	View transaction history after withdrawal	Withdrawal Transaction	Transaction displayed	Pass
TC13	View empty transaction history	No Transactions	No transaction found message display	Pass
TC14	Enter invalid menu choice	9	Invalid choice message displayed	Pass
TC15	Enter alphabet instead of amount	abc	Exception handled successfully	Pass
TC16	Enter special characters as amount	@#$	Exception handled successfully	Pass
TC17	Perform multiple deposits	Multiple Deposits	Balance updated correctly	Pass

TC18	Perform multiple withdrawals	Multiple Withdrawals	Balance updated correctly	Pass
TC19	Check balance after multiple transactions	Mixed Transactions	Correct balance displayed	Pass
TC20	Exit application	Exit Option	Program terminated successfully	Pass

10 Screenshot: -

1. Menu Options

```
PS C:\Users\Saniya Shaikh> & "C:/Users/Saniya Shaikh/AppData/Local/Programs
======================== ATM Simulation ========================
1. Deposit
2. Withdraw
3. Balance Inquiry
4. Transaction History
5. Exit
Enter your choice:
```

2. Enter choice and deposit amount

```
======================== ATM Simulation ========================
1. Deposit
2. Withdraw
3. Balance Inquiry
4. Transaction History
5. Exit
Enter your choice: 1
Enter amount to deposit: 14000
₹14000.0 deposited successfully.
Current Balance: ₹15000.0
```

3. Withdraw money

```
======================== ATM Simulation ============
1. Deposit
2. Withdraw
3. Balance Inquiry
4. Transaction History
5. Exit
Enter your choice: 2
Enter amount to withdraw: 5000
₹5000.0 withdrawn successfully.
Current Balance: ₹10000.0
```

4. Balance Inquiry

```
1. Deposit
2. Withdraw
3. Balance Inquiry
4. Transaction History
5. Exit
Enter your choice: 3

Current Balance: ₹10000.0
```

5. Transaction History

```
1. Deposit
2. Withdraw
3. Balance Inquiry
4. Transaction History
5. Exit
Enter your choice: 4

Transaction History
Deposited ₹14000.0
Withdrawn ₹5000.0
```

6. Exit

```
========================= ATM Simulation ====
1. Deposit
2. Withdraw
3. Balance Inquiry
4. Transaction History
5. Exit
Enter your choice: 5
Thank You for Using ATM.
```

11 Conclusion: -

The ATM Simulation System successfully demonstrates the implementation of Python programming concepts such as conditional statements, functions, loops, lists, and exception handling. The project provides practical experience in handling banking transactions and managing account balances through a menu-driven application. It serves as a strong foundation for developing more advanced financial and banking applications in the future.

12 **GitHub Repository Link: -**

https://github.com/saniya2698/ATM-Simulation-System-/tree/main